

DETAILED DESCRIPTION OF THE INVENTION

Technical Advantages and Differentiation

The penile anatomical geometry profiling system described herein provides several technical advantages over conventional manual measurement techniques and existing sizing tools. Traditional approaches relying on rulers, flexible tapes, or visual estimation capture limited scalar values and are susceptible to user error, inconsistent reference positioning, tissue compression variability, and lack of reproducibility. The disclosed system instead acquires synchronized radial distance and longitudinal displacement data, enabling reconstruction of a structured three-dimensional geometric model rather than isolated dimensional values.

A primary technical advantage lies in the integration of time-of-flight ranging with motion-indexed positional tracking. This synchronized acquisition architecture enables continuous surface sampling along a defined longitudinal axis while maintaining spatial indexing integrity. By correlating distance values with displacement measurements, the system reduces ambiguity associated with static, single-point measurement methods.

The incorporation of multi-sensor configurations, including multi-zone ToF arrays and redundant optical motion sensors, improves reconstruction fidelity and reduces sensitivity to localized signal noise. Data fusion routines and error-correction algorithms further enhance geometric accuracy compared to devices that rely solely on single-sensor proximity measurement.

Anti-spoofing mechanisms incorporating physiological verification sensors provide an additional layer of authenticity not present in conventional sizing tools. This capability supports measurement validation and reduces the likelihood of false or non-biological measurement inputs.

The generation of encoded, standardized geometry profiles allows structured storage, longitudinal comparison, normalization against population datasets, and compatibility with calibration or personalization engines. Unlike conventional measurement tools that provide transient numerical outputs, the disclosed system produces a reusable digital structural representation.

Security architecture including on-device encryption, secure boot mechanisms, and cryptographic authentication further differentiates the system by protecting sensitive anatomical data during acquisition and transmission.

Collectively, the synchronized sensing architecture, computational reconstruction framework, encoded profile generation, and security integration establish a technically differentiated platform for generating user-specific penile anatomical geometry profiles beyond the capabilities of traditional measurement approaches.